AP BIOLOGY TOPIC 7.11

Variations in Populations, Extinction & Adaptive Radiation

UNIT 7: NATURAL SELECTION (13–20% EXAM WEIGHT)

EVO-3.G | EVO-3.H | EVO-3.I | SYI-3.D | MASS EXTINCTIONS

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 7.11 (VARIATIONS IN POPULATIONS, EXTINCTION & ADAPTIVE RADIATION)

CED Core Vocabulary

High Genetic Diversity = Greater Resilience — More allelic variation increases the probability that at least some individuals express phenotypes capable of surviving a novel pathogen, toxin, or climate shift.

Genetic drift — reduces diversity in small populations, whereas gene flow and balancing selection (heterozygote advantage) help maintain allele diversity.

California Condors & Prairie Chickens — Severe bottlenecks caused inbreeding depression and reduced egg hatchability until rescued by genetic translocation (gene flow).

Balancing Selection — Preserves multiple alleles at higher frequencies in a gene pool via: Heterozygote Advantage: E.g., Sickle-cell allele (HbS) confers malaria resistance in...

Low Genetic Diversity = High Extinction Risk — Genetically uniform populations share the same vulnerabilities — a single novel disease or environmental change can wipe out the entire population.

Tasmanian Devils — Low MHC immune gene diversity allowed contagious Devil Facial Tumor Disease (DFTD) to spread without immune rejection.

Allele Fitness Depends on Environmental Context — An allele that is deleterious or neutral in one environment can become strongly advantageous if the environment changes.

Frequency-Dependent Selection — Fitness of a phenotype depends on how common it is (e.g., scale-eating cichlid fish mouth asymmetry).

§2. 1. GENETIC DIVERSITY & POPULATION RESILIENCE

Biological Insurance

- **Genetic Diversity = Variety of Alleles Within a Population:** Determines how well a population can respond to biotic or abiotic environmental shifts.
- **High Genetic Diversity = Greater Resilience:**
  - More allelic variation increases the probability that at least *some individuals* express phenotypes capable of surviving a novel pathogen, toxin, or climate shift.
  - Those survivors reproduce and allow the species to persist rather than go extinct.
- **Low Genetic Diversity = High Extinction Risk:**
  - Genetically uniform populations share the same vulnerabilities — a single novel disease or environmental change can wipe out the entire population.
  - **Genetic drift** reduces diversity in small populations, whereas **gene flow** and **balancing selection (heterozygote advantage)** help maintain allele diversity.

[MECHANISTIC NOTE] **Genetic diversity = biological insurance against environmental change.**

[COMMON AP EXAM FRQ TRAP] **Large population size ≠ high genetic diversity!** A population can have millions of individuals today (e.g., Northern elephant seals or monoculture crops) yet possess dangerously low genetic diversity due to a past bottleneck or clonal propagation!

[EXAMPLE & BIOLOGICAL APPLICATION] **High-Yield AP Case Studies of Low Genetic Diversity Vulnerability:**

- **Wild Cheetahs (*Acinonyx jubatus*):** Survived two severe historical bottlenecks → 99% genetically identical (can accept skin grafts from unrelated cheetahs!); highly vulnerable to feline coronavirus (FIP) and poor sperm quality.
- **Tasmanian Devils:** Low MHC immune gene diversity allowed contagious Devil Facial Tumor Disease (DFTD) to spread without immune rejection.
- **Cavendish Bananas & Irish Potato Famine (1840s):** Clonal asexual monocultures devastated by Panama disease fungus (TR4) and *Phytophthora infestans* water mold.
- **California Condors & Prairie Chickens:** Severe bottlenecks caused inbreeding depression and reduced egg hatchability until rescued by genetic translocation (gene flow).

§3. 2. CONTEXT-DEPENDENT ALLELE FITNESS & BALANCING SELECTION

Changing Pressures

- **Allele Fitness Depends on Environmental Context:** An allele that is deleterious or neutral in one environment can become strongly advantageous if the environment changes.
- **Balancing Selection:** Preserves multiple alleles at higher frequencies in a gene pool via:
  - **Heterozygote Advantage:** E.g., Sickle-cell allele (**HbS**) confers malaria resistance in **HbA/HbS** heterozygotes in sub-Saharan Africa, but is disadvantageous where malaria is absent.
  - **Frequency-Dependent Selection:** Fitness of a phenotype depends on how common it is (e.g., scale-eating cichlid fish mouth asymmetry).

§4. 3. EXTINCTION: BACKGROUND VS. MASS EXTINCTIONS

CED EVO-3.G – 3.J

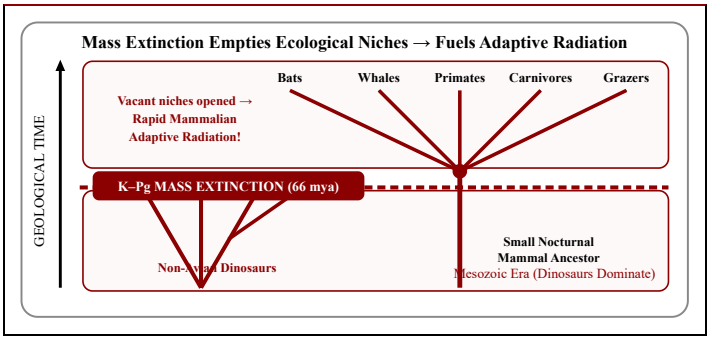
HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

3. Extinction ➔ **Why Extinction Occurs** ➔ **Background Extinction Rate**

- **Why Extinction Occurs:** Extinction happens when environmental conditions change faster than a species can adapt (insufficient genetic variation to match selective pressures). Over **99% of all species** that ever lived are now extinct!
- **Background Extinction Rate:** Slow, steady baseline rate of extinction resulting from normal ecological interactions (competition, predation, local habitat shifts).
- **5 Historical Mass Extinctions ("The Big Five"):** Global, rapid catastrophic events where **50%–96% of all species** were wiped out across Earth in a geologically short period:
  - **Permian-Triassic ("Great Dying," ~252 mya):** Massive Siberian volcanic eruptions, global warming, and ocean anoxia/acidification killed **~96% of marine species** and 70% of terrestrial vertebrates.
  - **Cretaceous-Paleogene (K–Pg / K–T, ~66 mya):** Chicxulub asteroid impact (iridium layer in clay) + Deccan Traps volcanism blocked sunlight, collapsing photosynthetic food webs and wiping out **non-avian dinosaurs** (~75% of all species).
- **6th Mass Extinction (Anthropocene / Holocene):** Currently driven by **human activities** — habitat destruction/fragmentation, climate change, ocean acidification, pollution, overexploitation, and invasive species — at **100 to 1,000× the background rate**.

§5. 4. ECOLOGICAL CONSEQUENCE: VACANT NICHES & ADAPTIVE RADIATION

EVO-3.I



- **Mass Extinctions Open Ecological Niches:** When dominant competitors/predators are eliminated, surviving lineages experience **ecological release** and rapidly diversify via **adaptive radiation** to fill the newly vacant niches (though recovery takes 5–10+ million years).

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 7.11 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 7.11: Variations in Populations, Extinction & Adaptive Radiation ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 1, 2 & 3 (Artificial Selection, Hardy-Weinberg Mathematical Modeling & BLAST Bioinformatics): Comparing allele frequency shifts across generations under directional selection vs. small-population genetic drift simulations, and constructing maximum-parsimony cladograms from BLAST percentage sequence identity matrices (Cytochrome c / β-globin). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Golden Rule: Hardy-Weinberg Equilibrium (HWE): Allele frequencies $p + q = 1$; Genotype frequencies p^2 (homozygous dominant) + $2pq$ (heterozygous carriers) + q^2 (homozygous recessive) = 1. Golden Calculation Sequence: ALWAYS begin with the recessive phenotype frequency (q^2), take $\sqrt{(q^2)}$ to find q, compute $p = 1 - q$, and then calculate $2pq$ (heterozygote carrier frequency). Never take the square root of the dominant phenotype frequency ($p^2 + 2pq$)! **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Evolutionary & Biomedical Disruption Case Study: Incomplete antibiotic regimens or agricultural antibiotic overuse exert strong directional selection on bacterial populations (e.g., MRSA or MDR-Tuberculosis): susceptible cells die while rare, pre-existing resistant mutants (arising via random mutation or R-plasmid conjugation) survive and reproduce. In conservation biology, severe population bottlenecks (e.g., Northern Elephant Seals, African Cheetahs) drastically erode genetic diversity via genetic drift, increasing homozygosity for deleterious recessive alleles and vulnerability to novel pathogens.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 7.11 to prove why the claim holds.

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